Reward function

$$r_t = -\left(w_1 x_{1,t} + w_2 x_{2,t} + w_3 x_{3,t} + w_4 + w_5 \left(1 - \cos \psi_t\right)\right) \tag{1}$$

where, for decision step t (10 s):

- $x_{1,t} \in \{0, 1\}$ is 1 if a new loss of separation (< 5 NM) started during the step, with $w_1 = 10$;
- $x_{2,t} \in \{0, 1\}$ is 1 if the aircraft newly entered a restricted area, with $w_2 = 10$;
- $x_{3,t} \in \{0, 1\}$ is 1 if the aircraft newly left the sector, with $w_3 = 10$;
- $w_4 = 5/300$ is a time penalty per decision, -5 over a full 3000 s episode;
- ψ_t is the angle between the aircraft's heading and the bearing to its goal at the end of the step, with $w_5 = 12/300$.

In training, a new violation ($x_{1,t}$, $x_{2,t}$ or $x_{3,t}$) also ends the episode. In evaluation the competition's rules apply and nothing ends early. The reward is never positive.
The weights balance two failures in PPO's discounted view ($\gamma = 0.98$, a horizon of about 50 decisions). Circling forever costs 2.8 against 0.7 for flying to the goal, so the agent goes even when that is up to 21% riskier. The worst way to keep flying costs 4.8, half of a violation, so ending the episode by a violation never pays.